

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; kimi-k3

Abstract

This report evaluates `moonshotai/kimi-k3` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron leads Cache hit rate in all routing modes; Infron leads Observed cost in all routing modes; Infron leads Throughput in 2/4 routing modes; OpenRouter leads in 2/4; 0/4 tie; Infron leads E2E latency in 2/4 routing modes; OpenRouter leads in 2/4; 0/4 tie; Infron leads Streaming TTFT in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie.

Overall, Infron's cross-mode strengths are Streaming TTFT, observed cost, cache reuse, while OpenRouter does not show a clear cross-mode strength in this run. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform–level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput Infron wins 2/4 Max advantage 40.12%, higher is better	Observed cost Infron wins 4/4 Max advantage 1.57%, lower is better	E2E latency Infron wins 2/4 Max advantage 40.10%, lower is better	Streaming TTFT Infron wins 3/4 Max advantage 59.95%, lower is better	Cache hit rate Infron wins 4/4 Max advantage 11.92%, higher is better
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Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	OpenRouter advantage 40.12%	Infron advantage 136.26%	OpenRouter advantage 40.10%	OpenRouter advantage 33.14%	Infron advantage 11.92%
Price First price	Infron advantage 28.89%	Infron advantage 147.41%	Infron advantage 28.90%	Infron advantage 59.95%	Infron advantage 4.90%
Latency First latency	Infron advantage 26.06%	Infron advantage 1.57%	Infron advantage 26.06%	Infron advantage 58.04%	Infron advantage 6.99%
TTFT First ttft	OpenRouter advantage 7.28%	Infron advantage 118.90%	OpenRouter advantage 7.26%	Infron advantage 15.05%	Infron advantage 2.76%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second <code>usage.prompt_tokens</code> deltas are limited to 50 tokens within each <code>sort/group/round</code> pair.	Methods and data quality
Cache reuse	Infron leads Cache hit rate in all routing modes	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in all routing modes	Overall metrics and provider drill-down
Performance	Infron leads Throughput in 2/4 routing modes; OpenRouter leads in 2/4; 0/4 tie; Infron leads E2E latency in 2/4 routing modes; OpenRouter leads in 2/4; 0/4 tie; Infron leads Streaming TTFT in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	Infron	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	Infron	Infron	Infron	Infron	Infron	Infron leads overall (5/5 metrics)
Latency First	Minimize full–response waiting time	Infron	Infron	Infron	Infron	Infron	Infron leads overall (5/5 metrics)
TTFT First	Minimize streaming first–token time	Infron	Infron	OpenRouter	OpenRouter	Infron	Infron leads overall (3/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference–platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first–token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token–level cache hit rate for repeated stable prefixes.	Second–request cache–read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response–returned `usage.prompt_tokens` as the input–token control while allowing small cross–platform accounting variance up to 50 tokens.
- Extends prompt–caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

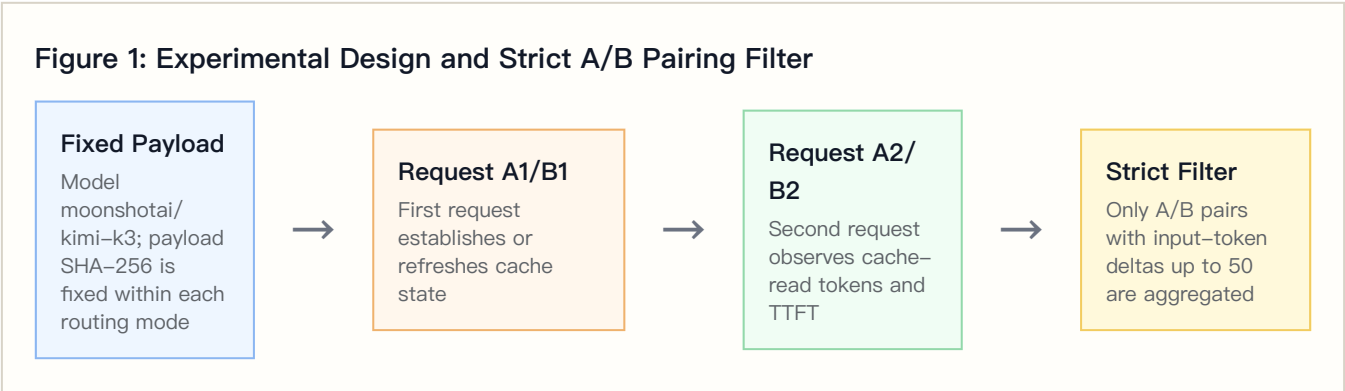
2. Experimental Design, Dataset, and Controls

2.1 Dataset Construction

The dataset is `controlled_cache_probe`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with <code>cache_read_tokens > 0</code>	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	moonshotai/kimi-k3
Provider model IDs	infron: moonshotai/kimi-k3 ; openrouter: moonshotai/kimi-k3
Platforms	Infron and OpenRouter
API protocol	/v1/chat/completions
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	short ≈1500, medium ≈8000, long ≈32000
Excluded records	0

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	200	6295313	90.73%	\$3.68728300	1.17 tok/s	13624.80 ms	12485.09 ms
Throughput First	OpenRouter	200	6295316	81.07%	\$8.71151400	1.65 tok/s	9724.88 ms	9377.65 ms
Price First	Infron	200	6295316	96.95%	\$2.00315200	3.21 tok/s	4981.17 ms	3664.40 ms
Price First	OpenRouter	200	6295283	92.42%	\$4.95605220	2.49 tok/s	6420.97 ms	5861.22 ms
Latency First	Infron	200	6295316	96.95%	\$2.00315200	3.38 tok/s	4731.76 ms	3447.95 ms
Latency First	OpenRouter	200	6295286	90.61%	\$5.15808300	2.68 tok/s	5965.07 ms	5448.98 ms
TTFT First	Infron	200	6295316	96.95%	\$2.00315200	2.75 tok/s	5825.79 ms	4439.04 ms
TTFT First	OpenRouter	200	6295292	94.34%	\$4.38489300	2.95 tok/s	5431.57 ms	5107.21 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	9800.67 ms	36846.55 ms	56439.24 ms	8248.98 ms	36131.33 ms	56229.21 ms
Throughput First	OpenRouter	6530.30 ms	26234.09 ms	32973.48 ms	6125.86 ms	25952.04 ms	32686.90 ms
Price First	Infron	4823.16 ms	6146.20 ms	8472.32 ms	3663.95 ms	5220.94 ms	7039.72 ms
Price First	OpenRouter	4890.12 ms	15134.51 ms	25891.55 ms	4544.51 ms	14249.28 ms	25542.00 ms
Latency First	Infron	4648.21 ms	6312.50 ms	7643.37 ms	3455.07 ms	4804.87 ms	5783.06 ms
Latency First	OpenRouter	4997.66 ms	11876.76 ms	22451.02 ms	4670.13 ms	11081.37 ms	19503.72 ms
TTFT First	Infron	4953.04 ms	10094.64 ms	10373.94 ms	3846.52 ms	9475.50 ms	9759.26 ms
TTFT First	OpenRouter	4812.98 ms	10981.30 ms	13986.84 ms	4519.13 ms	10346.70 ms	13775.03 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	–7799.83 ms	–10338.09 ms to –5296.81 ms	<0.001	200	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	–6214.88 ms	–8704.52 ms to –3715.99 ms	<0.001	200	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	–0.5647 tok/s	–0.7766 tok/s to –0.3486 tok/s	<0.001	200	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	\$0.02512115	\$0.01921239 to \$0.03120413	<0.001	200	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	17.72 pp	11.54 pp to 23.99 pp	<0.001	200	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	2879.60 ms	1915.71 ms to 3906.56 ms	<0.001	200	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	4393.65 ms	3516.24 ms to 5373.81 ms	<0.001	200	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	0.2429 tok/s	0.0812 tok/s to 0.3990 tok/s	0.0042	200	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	\$0.01476450	\$0.01308828 to \$0.01654511	<0.001	200	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	8.39 pp	5.43 pp to 11.72 pp	<0.001	200	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	2466.62 ms	1716.63 ms to 3336.59 ms	<0.001	200	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	4002.05 ms	3434.36 ms to 4615.06 ms	<0.001	200	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	0.3603 tok/s	0.2238 tok/s to 0.4957 tok/s	<0.001	200	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	\$0.01577465	\$0.01372813 to \$0.01788192	<0.001	200	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	10.80 pp	7.03 pp to 15.25 pp	<0.001	200	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	-788.43 ms	-1468.94 ms to -150.57 ms	0.0245	200	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	1336.33 ms	716.99 ms to 1945.41 ms	<0.001	200	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	-0.3480 tok/s	-0.5364 tok/s to -0.1692 tok/s	0.0013	200	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	\$0.01190870	\$0.01025233 to \$0.01376262	<0.001	200	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	4.44 pp	2.05 pp to 7.22 pp	<0.001	200	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	6221	15.5525	400	12508.15 ms	12485.09 ms	13624.80 ms
Throughput First	OpenRouter	6400	16.0000	400	9377.65 ms	9377.65 ms	9724.88 ms
Price First	Infron	6238	15.5950	400	3696.71 ms	3664.40 ms	4981.17 ms
Price First	OpenRouter	6382	15.9550	400	5861.22 ms	5861.22 ms	6420.97 ms
Latency First	Infron	6213	15.5325	400	3458.50 ms	3447.95 ms	4731.76 ms
Latency First	OpenRouter	6388	15.9700	400	5448.98 ms	5448.98 ms	5965.07 ms
TTFT First	Infron	6279	15.6975	400	4464.31 ms	4439.04 ms	5825.79 ms
TTFT First	OpenRouter	6388	15.9700	400	5107.21 ms	5107.21 ms	5431.57 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions` ; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
chat_completions	/v1/chat/completions	Infron	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%
chat_completions	/v1/chat/completions	OpenRouter	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

**Figure 3-E2:
Normalized
capability
contour**

Five metrics are normalized to 0-100, where higher is better.

**Figure 3-E3:
Cost-cache
efficiency plane**

X-axis is token cache hit rate; Y-axis is observed total cost.

**Figure 3-E1:
Routing-mode
core
metric
comparison**

Infron and OpenRouter are shown side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache–domain and performance differences.

Routing–Mode Drill–Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

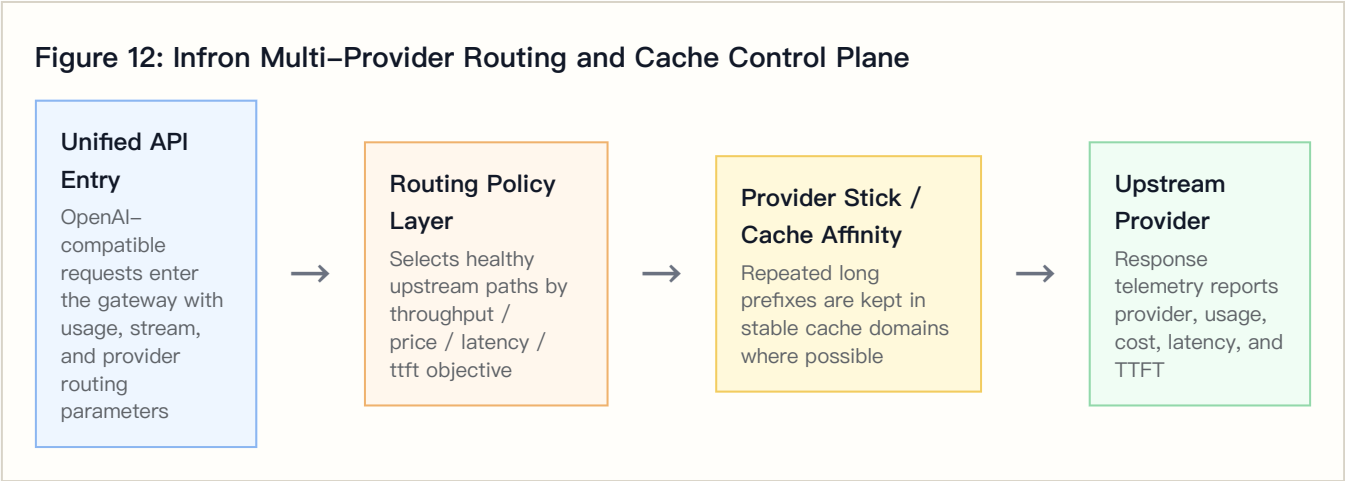
Figure 4–E3: Latency First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	400	400	<code>infron/us</code> 382, <code>fireworks</code> 17, <code>moonshotai</code> 1

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	OpenRouter	400	400	Fireworks 400
Price First	Infron	400	400	infron/us 400
Price First	OpenRouter	400	400	Fireworks 358, DigitalOcean 31, Moonshot AI 6, Together 5
Latency First	Infron	400	400	infron/us 400
Latency First	OpenRouter	400	400	Fireworks 376, DigitalOcean 14, Together 6, Moonshot AI 4
TTFT First	Infron	400	400	infron/us 400
TTFT First	OpenRouter	400	400	Fireworks 380, DigitalOcean 12, Moonshot AI 4, Together 4

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Throughput First	Infron	infron/us	382	95.50%	196/186	196	11753.86 ms	12907.71 ms	5804839	6112
Throughput First	Infron	fireworks	17	4.25%	4/13	14	28760.21 ms	29649.01 ms	454027	272
Throughput First	Infron	moonshotai	1	0.25%	0/1	1	15136.47 ms	15139.59 ms	36447	16
Throughput First	OpenRouter	Fireworks	400	100.00%	200/200	200	9377.65 ms	9724.88 ms	6295316	6400
Price First	Infron	infron/us	400	100.00%	200/200	200	3664.40 ms	4981.17 ms	6295316	6400
Price First	OpenRouter	Fireworks	358	89.50%	182/176	182	5524.65 ms	5903.06 ms	5922694	5728
Price First	OpenRouter	DigitalOcean	31	7.75%	15/16	18	6537.15 ms	9321.31 ms	278541	496
Price First	OpenRouter	Moonshot AI	6	1.50%	1/5	5	18651.48 ms	18820.47 ms	47948	96
Price First	OpenRouter	Together	5	1.25%	2/3	3	10420.27 ms	10641.48 ms	46100	80
Latency First	Infron	infron/us	400	100.00%	200/200	200	3447.95 ms	4731.76 ms	6295316	6400
Latency First	OpenRouter	Fireworks	376	94.00%	186/190	190	5444.63 ms	5781.67 ms	6081336	6016

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Latency First	OpenRouter	DigitalOcean	14	3.50%	8/6	8	5039.73 ms	10393.45 ms	129122	224
Latency First	OpenRouter	Together	6	1.50%	4/2	4	6672.54 ms	7117.37 ms	47948	96
Latency First	OpenRouter	Moonshot AI	4	1.00%	2/2	2	5454.52 ms	5977.31 ms	36880	64
TTFT First	Infron	infron/us	400	100.00%	200/200	200	4439.04 ms	5825.79 ms	6295316	6400
TTFT First	OpenRouter	Fireworks	380	95.00%	190/190	190	5163.33 ms	5473.74 ms	6110856	6080
TTFT First	OpenRouter	DigitalOcean	12	3.00%	6/6	6	3362.64 ms	3993.35 ms	110676	192
TTFT First	OpenRouter	Moonshot AI	4	1.00%	2/2	2	6561.41 ms	6669.14 ms	36880	64
TTFT First	OpenRouter	Together	4	1.00%	2/2	2	3555.57 ms	4502.63 ms	36880	64

7.1 Cache–Rate and Cost Divergence Drill–Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing–mode differences.

Routing mode	Cache–hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	+9.66 pp	0.42x	infron/us 95.50%	Fireworks 100.00%	–179	Infron leads both cache and cost
Price First	+4.53 pp	0.40x	infron/us 100.00%	Fireworks 89.50%	–144	Infron leads both cache and cost
Latency First	+6.33 pp	0.39x	infron/us 100.00%	Fireworks 94.00%	–175	Infron leads both cache and cost
TTFT First	+2.60 pp	0.46x	infron/us 100.00%	Fireworks 95.00%	–109	Infron leads both cache and cost

8. Stratified Results: Prompt–Length Cache Performance

This section aggregates second–request cache–read tokens, token–level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt–length tier. Bold cells mark the advantaged side within each tier.

Prompt–Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	Infron	268	496068	270336	54.50%	\$1.31663000	6135.05 ms	4464.07 ms
short	1500	OpenRouter	268	496065	268288	54.08%	\$2.44459830	5090.25 ms	4719.55 ms
medium	8000	Infron	268	2471764	2457600	99.43%	\$1.82740800	6462.67 ms	5275.87 ms
medium	8000	OpenRouter	268	2471719	1695888	68.61%	\$9.15939990	6720.31 ms	6127.56 ms
long	32000	Infron	264	9622797	9282560	96.46%	\$6.55270100	9304.97 ms	8321.94 ms
long	32000	OpenRouter	264	9622800	9318400	96.84%	\$11.60654400	8876.02 ms	8530.26 ms

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
short	Throughput First	52.02%	52.84%	OpenRouter
short	Price First	55.32%	52.85%	Infron
short	Latency First	55.32%	55.32%	tie
short	TTFT First	55.32%	55.32%	tie
medium	Throughput First	97.94%	42.75%	Infron
medium	Price First	99.92%	77.35%	Infron
medium	Latency First	99.92%	67.67%	Infron
medium	TTFT First	99.92%	86.67%	Infron
long	Throughput First	90.88%	92.37%	OpenRouter
long	Price First	98.33%	98.33%	tie
long	Latency First	98.33%	98.33%	tie
long	TTFT First	98.33%	98.33%	tie

9. Stratified Results: Group–Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	96.50%	\$0.90213000	26773.70 ms	25369.36 ms
Infron	2	50	50	97.06%	\$0.62141700	32964.66 ms	30848.74 ms
Infron	3	50	50	96.93%	\$0.61478600	43404.76 ms	43153.89 ms
Infron	4	50	50	71.94%	\$1.54895000	36231.21 ms	35598.26 ms
OpenRouter	1	50	50	82.56%	\$2.59553160	27787.99 ms	27440.76 ms
OpenRouter	2	50	50	85.62%	\$1.96136100	28482.92 ms	28332.07 ms
OpenRouter	3	50	50	78.46%	\$2.26411380	22323.94 ms	22063.26 ms
OpenRouter	4	50	50	77.52%	\$1.89050760	7973.88 ms	7755.94 ms

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	96.90%	\$0.49295200	6802.70 ms	4873.53 ms
Infron	2	50	50	97.06%	\$0.50862200	6529.45 ms	5250.96 ms
Infron	3	50	50	96.93%	\$0.50862600	6046.81 ms	5542.86 ms
Infron	4	50	50	96.90%	\$0.49295200	5450.69 ms	4673.38 ms
OpenRouter	1	50	50	91.55%	\$1.24684650	11695.15 ms	11359.45 ms
OpenRouter	2	50	50	94.23%	\$1.20273570	17563.09 ms	15088.68 ms
OpenRouter	3	50	50	92.09%	\$1.32622380	12287.41 ms	11894.17 ms
OpenRouter	4	50	50	91.72%	\$1.18024620	20289.79 ms	16462.30 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	96.90%	\$0.49295200	6373.04 ms	4712.09 ms
Infron	2	50	50	97.06%	\$0.50862200	5691.79 ms	4814.83 ms
Infron	3	50	50	96.93%	\$0.50862600	6446.67 ms	4787.97 ms
Infron	4	50	50	96.90%	\$0.49295200	6431.59 ms	4819.55 ms
OpenRouter	1	50	50	89.60%	\$1.35305370	10053.82 ms	9500.33 ms
OpenRouter	2	50	50	89.60%	\$1.29851460	11718.37 ms	10785.13 ms
OpenRouter	3	50	50	92.31%	\$1.24726680	9381.92 ms	8847.29 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
OpenRouter	4	50	50	90.93%	\$1.25924790	13189.93 ms	13020.56 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	96.90%	\$0.49295200	10270.40 ms	9751.20 ms
Infron	2	50	50	97.06%	\$0.50862200	10201.98 ms	9472.41 ms
Infron	3	50	50	96.93%	\$0.50862600	5754.34 ms	4722.51 ms
Infron	4	50	50	96.90%	\$0.49295200	6828.01 ms	5544.89 ms
OpenRouter	1	50	50	93.31%	\$1.16395980	12118.73 ms	12053.78 ms
OpenRouter	2	50	50	95.53%	\$1.01721720	12599.75 ms	12427.92 ms
OpenRouter	3	50	50	93.47%	\$1.20164760	10260.17 ms	10052.08 ms
OpenRouter	4	50	50	95.04%	\$1.00206840	9435.39 ms	9126.06 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	OpenRouter leads overall (3/5 metrics)	Batch generation, offline summaries, backend processing	Cache and cost are stronger; still check speed SLA
Price First	Minimize request and token cost	Infron leads overall (5/5 metrics)	High-frequency templates, support automation, marketing, RAG prefixes	Cache and cost are stronger; still check speed SLA
Latency First	Minimize full-response waiting time	Infron leads overall (5/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	Cache and cost are stronger; still check speed SLA
TTFT First	Minimize streaming first-token time	Infron leads overall (3/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	Cache and cost are stronger; still check speed SLA

11. Conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	Infron	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	Infron	Infron	Infron	Infron	Infron	Infron leads overall (5/5 metrics)
Latency First	Minimize full–response waiting time	Infron	Infron	Infron	Infron	Infron	Infron leads overall (5/5 metrics)
TTFT First	Minimize streaming first–token time	Infron	Infron	OpenRouter	OpenRouter	Infron	Infron leads overall (3/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short–window stability but not day–level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built–in templates do not cover every workload distribution	Use sanitized production–stratified corpora	Discuss representative long–context templates only
Cost–field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	export/kimi_k3_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions summary.json
Paired dataset	export/kimi_k3_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_pairs.csv
Request–level dataset	export/kimi_k3_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_requests.jsonl

Artifact	Path
Filtered structured records	<code>export/kimi_k3_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_records.json</code>
Excluded-record audit	<code>export/kimi_k3_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_records_excluded.json</code>
Test source	<code>tests/test_rerun_routing_sort_cache_cost_ab.py</code>
Benchmark runner source	<code>scripts/rerun_routing_sort_cache_cost_ab.py</code>
HTML report renderer source	<code>scripts/render_glm52_deepseek_style_report.py</code>
Dataset reference	<code>business_representative</code> built-in representative business templates; request-level export is <code>benchmark_requests</code>